popquiz dynamic equilibrium, static friction, newtons I

This is a preview of the published version of the quiz

Started: Apr 29 at 7:58pm

Quiz Instructions

bonus

Question 1 14.5 pts

Here is a dynamic equilibrium. The sled is the system and moves at a constant velocity. The sled is 1kg. The kid pulls at a 40 degrees angle above the horizontal. The pull is 10N. What is the kinetic friction?

hint: work along the x-axis. x-component of the pull = kinetic friction (no acceleration)

how to find the x-component of the pull?

- about 10N
- There is no kinetic friction
- about 8N
- about 6N

Question 2 14.5 pts

A truck and a mosquito collide on the high way. What is true.

- they both epxerience the same force
- the truck experiences the largest force
- can not be determined
- the mousquito experiences the largest force

Question 3 5 pts

bonus. Building on the previous question. Which one will experience the largest acceleration. (change in state of motion)

- truck
- mosquito
- can no be determined
- they experience the same acceleration

Question 4 14.5 pts

https://phet.colorado.edu/sims/html/forces-and-motion-basics/latest/forces-and-motion-basics_en.html (https://phet.colorado.edu/sims/html/forces-and-motion-basics/latest/forces-and-motion-basics_en.html)

Use the app above. Go to the tab friction. Click all the squares (upper right).

HAve the dummy push on the 50kg crate slowly until it starts to move. slowly

What is the maximum static friction reaches before kinetic friction takes over

- about 125N
- about 94N
- about 200N
- about 500N

Question 5 14.5 pts

Building on the previous question. There is the relationship between the magnitude of fs (static friction) and the magnitude of N (normal force):

(Static friction max)= us x (Normal force)

What is the us (coeff of friction) in our example? take g=10m/s/s

hint: you have fs, find the normal force. (up=down so normal force = weight = mg)

- 1
- 0.188
- 0.25
- 0.4

Question 6 15 pts

https://phet.colorado.edu/sims/html/forces-and-motion-basics/latest/forces-and-motion-basics_all.html (https://phet.colorado.edu/sims/html/forces-and-motion-basics/latest/forces-and-motion-basics_all.html)

Have the dummy push the crate so it moves. What is the kinetic frictional force?

- about 90N
- about 120N
- about 50N
- about 10N

Question 7 15 pts

The Dummy take the 50kg crate on the Moon. The friction is the same. He pushes on the crate. Is it easier on the Moon?

(less force for same acceleration?)

- yes (less weight)
- I missed the class

○ no (same inertia)

○ I depends on the color of the dummy

Question 8 15 pts

If you weigh 150 pounds (earth pulls on you with 150lbs) . what is the reaction force?

hint: the pair is you/earth so only you and earth. no menage a 3.

○ you are pulling on earth with 150 lbs

○ you are pulling on the ground

○ you pull on the ground.

○ the ground pulls on you

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